

SUPPLEMENTAL MATERIAL

Data S1.

Supplemental Methods

Eligibility and Recruitment/Screening Process

Eligibility criteria were:

- 18 to 35 years old at the time of recruitment;
- never smokers;
- body mass index between 19 to 28 kg/m² with body weight greater than 50 kg;
- no history of heart disease, diabetes, kidney disease, systemic sclerosis, or any chronic inflammatory disease such as asthma, arthritis, or severe allergies;
- normal non-hypertensive blood pressure, normal electrocardiogram, spirometry values greater than 70% of the predicted value for the age/gender, and normal blood test results (including no evidence of iron-deficient anemia), as determined at the screening exam;
- not currently taking statins, anti-inflammatory medication, or other medications unless cleared by the study physician during the screening exam (cleared medications: oral contraceptives, some daily anti-depression/anxiety medications);
- no use of tetrahydrocannabinol or illicit drugs within the past three months;
- no ear or abdominal/thoracic surgery in the past month; no cancer (current or in remission for less than six months); no central intravenous line or port; never had a mastectomy;
- no pacemaker;
- not currently pregnant, breastfeeding, or planning a pregnancy within six months;
- not regularly exposed to smoke, dust, fumes, or solvents (occupationally or recreationally/at home), or regularly burned candles or incense within the last three months;

- no history of claustrophobia;
- no fear of needles;
- not planning to donate blood during the timeframe of participation;
- no latex allergy; and
- live within 20 miles of the study facility and not planning to move more than 20 miles away within six months.

Individuals interested in participating in the study completed a screening questionnaire to determine potential eligibility. Individuals who appeared to meet criteria based on this questionnaire were asked to attend an in-person screening exam to ensure that the participant met health-based eligibility criteria. At the screening exam, medical staff measured the individual's height, weight, and blood pressure. A physician reviewed the recruitment questionnaire with the individual, conducted a physical exam, and reviewed the individual's medical history, including family history of cardiovascular and respiratory disease. Additionally, the individual performed an electrocardiogram, spirometry test, and a blood draw for analysis of complete blood count, comprehensive metabolic panel, lipid levels, and serum ferritin. A physician reviewed all results from the screening exam to make a final determination of eligibility for the study.

Additionally, individuals received a tour of the study facility and an overview of the study process, requirements, and expectations at the time of their screening exam. The goal of the tour and overview was to familiarize potential participants with the exposure facility, protocols, and staff, to reduce drop-out rates and alleviate potential stress-related reactions to the contrived exposure experience during their participation. During the tour, study staff showed individuals examples of cookstoves such as those used for the treatments, explained how the exposure chamber worked and safety features of the chamber (e.g., non-locking doors, the intercom and messaging

system to communicate with staff, emergency shut-offs, and pollution level monitoring), and described what the participant experience was like during exposure periods and throughout the health measurements. Individuals were able to see the exposure chamber and, if not in use during the time of their screening exam, enter the exposure chamber.

Treatment Sequence Assignments

The study protocol called for each participant to receive six exposure treatments (a clean air control and five treatments of cookstove air pollution). There was a washout period between treatments that was typically two weeks but up to six weeks by design, due to holidays and other similar schedule constraints. Our study followed a modified Williams square design, which is a Latin square crossover design that is balanced across treatments and first-order carry-over effects. We specified six unique sequences of treatments. Each sequence contained all six treatments administered in a unique order across the participant's six study sessions. Across all sequence groups, each treatment appeared once in each of the six "periods" (visit numbers) of the treatment orders (e.g., first through last assigned study session) and was both followed by and preceded by each other treatment exactly once. This design controls for time-invariant personal level factors and is robust to time-variant factors that might differ from one study session to the next. Participants were blinded to their sequence.

We conducted the study in three rounds (October 2016 to February 2017; March to June 2017; August 2017 to January 2018). In each round, two sequence groups (8 participants each) completed their full set of six treatments on alternating weeks. Additionally, within each sequence group, we divided participants into two subgroups (4 participants each) who completed their session on different days of the week (Mondays or Wednesdays). For the four participants who completed their study sessions on the same day, we staggered start times by 30 minutes.

Participants who missed a scheduled study session (due to illness or unforeseen conflict) were allowed to make up the missed treatment at the end of the sequence.

Participants were scheduled each study day with 30-minute staggered start times between 7:30AM and 9AM; each participant was scheduled for the same start time for each of their six study sessions. Assignment of participants into sequence groups, week days, and time slots was random; however, we did consider each participant's availability for aspects of the randomization (i.e., participants were recruited on a rolling basis into the ongoing study round and were allowed to specify whether they were not available to be placed into certain dates/time slots). Researchers were blind to the treatment orders within each sequence during the assignment process.

Participants who missed a scheduled study session (due to illness or other unforeseen conflict) were allowed to make up the missed session at the end of the sequence. Makeups were not necessarily completed on the same day of week or starting time as their regular schedule. Makeups were conducted ten days to 14 weeks after the last scheduled treatment, scheduled based upon participant and study schedule constraints. Participants remained blind to the treatment during makeups.

Treatments and Administration

Stove makes/models were as follows:

1. Liquefied petroleum gas [LPG] stove: Classic Single Burner 25000 BTU, WokSmith, China
2. Gasifier: Ace 1 Gasifier, African Clean Energy (Pty) Ltd, Lesotho
3. Forced draft (fan-powered) rocket elbow: HomeStove, Biolite, USA
4. Natural draft rocket elbow: G3300, Envirofit International, USA
5. Traditional three stone fire: open fire, bricks in U-shape used to contain fuel

The filtered air (target PM_{2.5}: 0 µg/m³) was generated by drawing conditioned laboratory air through a high-efficiency particulate air (HEPA) filter. Pollution was generated within a total-capture fume hood, diluted with HEPA-filtered laboratory air, and then drawn into the exposure chamber. A nephelometer (DustTrak DRX 8533, TSI Incorporated, USA) with a PM_{2.5} size-selective cyclone inlet was calibrated to the wood and LPG stoves separately (based on gravimetric filter data). The DustTrak and a gas analyzer (Siemens Ultramat 6E, Siemens AG, Germany) were used to monitor PM_{2.5}, carbon monoxide (CO), and oxygen levels in the chamber in real time; humidity and temperature were also monitored (Omega HX94BC transmitter and Type K thermocouple, OMEGA Engineering, USA). A dynamic control system (LabVIEW™, v15.0 32-bit, National Instruments, USA) was used to automate the flows of both dilution of pollution air based on real-time PM_{2.5} data received from the DustTrak.

While in the exposure chamber, participants were asked to remain seated at assigned desks and to avoid watching suspenseful videos, talking to each other, or talking on a cell phone for the duration of the exposure period; however, activities within the facility were not otherwise restricted (participants were allowed to use computers/internet, read books, listen to music, nap, etc.). Participants wore noise-canceling headphones while inside the exposure chamber, which reduced the noise generated by the exposure delivery system and allowed the study nurse to communicate with them from outside the chamber via intercom. We attempted to blind participants to their treatments on a given study day; however, full blinding was not feasible as higher-PM treatment levels (e.g., fan rocket, rocket elbow, three stone fire) have a distinct wood smoke smell that participants could identify when they entered the exposure chamber.

Additional characterization of stove emissions within the facility was conducted at the end of the study, at least twice per treatment type in randomized order. For each characterization test,

the facility was operated for two hours under the same conditions as during human exposures, but without participants present, on at least two occasions per treatment type. Air was sampled at breathing zone height 1 m from the wall of the facility (approximating the location where participants sat). Gravimetric PM_{2.5} measurements were made by sampling air at 16.7 L/min through a PM_{2.5} size-selective cyclone (URG Corporation, USA) onto polytetrafluoroethylene membrane filters (47mm, Tisch Environmental, USA), which were analyzed offline. Particle number size distributions (10 to 500 nm) were measured with a scanning mobility particle sizer (SMPS 3081-3785, TSI Inc., USA). Elemental and organic carbon concentrations were determined using thermo-optical analysis (OCEC Analyzer, Sunset Laboratory, USA) on pre-baked quartz filters (Tissuequartz, Pall Life Sciences, USA). The quartz filter samples were collected by sampling at 16.7 L/min through the PM_{2.5} size-selective cyclone. Nitrogen oxide and nitrogen dioxide were measured at 1 Hz via chemiluminescence (Model 42i-TL, Thermo Scientific, USA). Gas-phase carbonyls were measured by sampling onto 2,4-dinitrophenylhydrazine silica-based cartridges downstream of an ozone scrubber (Sep-Pak, Waters, USA) and analyzed offline using high-performance liquid chromatography with ultraviolet detection. Whole air samples were collected in 2-liter electropolished stainless steel canisters equipped with Silonite-coated flow controller valves (Entech Instruments Inc., USA) and analyzed offline for volatile organic compounds using gas chromatography/mass spectrometry.

Health Measurements

A series of cardiovascular and pulmonary health measurements was taken at four time points: before exposure, immediately after exposure, 3 hours post exposure, and 24 hours post exposure. Each series of measurements took approximately one hour to complete. Measurements were taken in the same order at each time point and across each study session, as indicated below:

1. Apply Holter monitor for heart rate variability (HRV) measurement (at the baseline time point only; the Holter monitor remained in place for the first three time points only).
2. Rest period in supine position lasting twenty minutes during time points one, two, and three (HRV data collected during last 10 minutes) and ten minutes during time point 4 (no HRV data collected).
3. Blood pressure and pulse wave analysis (for Augmentation Index) using SphygmoCor device (participant remained in supine position; measurements conducted on left side using appropriately sized cuffs).
4. Pulse wave velocity using SphygmoCor device (participant remained in supine position).
5. Spirometry using Easy-on device (participant in seated position).
6. Venous blood draw for analysis of inflammatory markers, complete blood count, and lipid levels.

Safety Considerations and Informed Consent

We received approval to conduct this study from the funding agency (NIH) and the Colorado State University Institutional Review Board. As part of this process, we provided both agencies with extensive justifications related to participant safety.

Only one of the study treatments (the traditional open fire wood stove; $500 \mu\text{g}/\text{m}^3$) resulted in exposure levels above the US daily regulatory standard for fine particulate matter (currently $35 \mu\text{g}/\text{m}^3$ for the 24-hour average); assuming $500 \mu\text{g}/\text{m}^3$ exposure for 2 hours and $8 \mu\text{g}/\text{m}^3$ for the rest of the 24 hours in a day (typical background concentration in Fort Collins, Colorado, the study location) would result in a 24-hour average of $49 \mu\text{g}/\text{m}^3$. Using similar calculations, the rest of the study treatments result in daily exposure averages below the daily standard for particulate matter

(and below the standards for other ambient criteria pollutants). Furthermore, the US Environmental Protection Agency National Ambient Air Quality Standards are set to protect all citizens, including those who are most susceptible (e.g., children, the elderly, or those with existing cardiovascular or respiratory disease). Our study included only healthy, never-smoker adults to minimize potential risks.

We used strict eligibility criteria to ensure participants in this study were healthy and therefore, at low risk of both acute and chronic harm from exposure to the air pollution in the study. We excluded potential participants who: had a history of heart disease, diabetes, or any chronic inflammatory disease (such as asthma, arthritis, or severe allergies); body mass index (BMI) less than 19 or greater than 25; were currently pregnant or planning a pregnancy during the exposure period; regularly taking statins or other anti-inflammatory medication; occupationally exposed to dust, fumes, solvents, or secondhand smoke. All participants were examined by a board-certified cardiologist prior to enrollment to determine eligibility, including a full medical history and routine physical exam, EKG, spirometry, and bloodwork. Normal EKG, spirometry, and blood work were prerequisites for entry into the study. Participants were instructed to tell study staff if they had any change in health during their participation in the study, so that we could reassess eligibility.

The exposure chamber facility was monitored in real-time by trained staff to ensure $PM_{2.5}$ concentrations were maintained within reasonable ranges of the target, planned concentrations and that CO, oxygen, temperature, and humidity remained within acceptable levels. The facility was equipped with an alarm that would trigger if the PM concentration delivered to the chamber exceeds $600 \mu g/m^3$ across a 2-min average, at which point the cookstove exhaust would automatically shut off and subjects were to be instructed to leave the chamber immediately (this

did not ever occur). Participants were informed of additional safety features, including non-locking doors that could easily be pushed open at any time, windows through which they could see study staff (and were being watched by study staff, including a registered nurse, at all times), monitoring of blood pressure and pulse oxygen every 15 minutes, and various methods for communicating with staff who were outside the facility. A physician and registered nurse were present during the controlled exposure periods; the physician remained on-call for 24 hours after the end of the exposure period.

The informed consent document contained a description of each of the exposures, comparing the level to easy-to-understand references (e.g., for $250 \mu\text{g}/\text{m}^3$ $\text{PM}_{2.5}$, we noted that similar exposures occurred in Fort Collins during the 2012 High Park fire, a major wildfire that occurred in a nearby area). The consent form described the risks of air pollution exposures, with following statements:

- Short-term exposure to air pollution ($\text{PM}_{2.5}$), like that which will be experienced in this study, poses limited risks to young, healthy participants. Long-term (over the course of decades) exposure to air pollution may increase the chance of adverse health effects, but this study is considered short-term exposure.
- Short-term exposure to carbon monoxide (CO) has been associated with headaches, dizziness, nausea, and/or vomiting. With the exception of headaches, these symptoms typically occur at concentrations higher than the maximum concentration (200 ppm) participants may experience while inside the SET facility. Participants will immediately be escorted out of the SET facility should CO concentrations in the SET facility exceed the recommended exposure limit (200 ppm) set forth by the National Institute for Occupational Safety and Health.

All participants went through the informed consent document in-person with study staff, who verbally explained the details in the written document to ensure participants understood the types of exposures and exposure levels. Participants were given the chance to ask questions.

Statistical Analysis: Confounders

Descriptive statistics were calculated and bivariate analyses were conducted to confirm that associations between these covariates and the treatment groups did not occur by chance or due to imbalances caused by missing data.

For categorical variables (e.g., yes/no consumption of alcohol; bike, car, walk, bus mode of commute), we tallied the number of individuals who reported each category by treatment type. We also tallied the number of individuals who changed their reported answer from one study period to the next. For continuous variables (e.g., ambient CO concentrations), we calculated the mean, minimum, and maximum value per treatment type. Descriptive analyses for continuous variables included determining mean, minimum, and maximum values for the variable by treatment type.

For variables that showed variation across treatments, we ran models to predict the confounder status by treatment type, to determine if there were meaningful differences in the variable by treatment type. The models predicted the outcome of the confounder status (e.g., binary alcohol, caffeine, and medication consumption status; continuous ambient conditions) with a fixed effect of categorical treatment type and a random person intercept to account for non-independence across repeated measures within our crossover design (i.e., each individual completing multiple treatments). We also ran a multivariable model that contained the treatment effect adjusted for multiple potential confounders.

Data S2.

Supplemental Results

Treatments

Summaries of the addition pollutant characterizations are provided in Table S1 and Figure S1. Gravimetric measurements and time-resolved PM_{2.5} measurements made during the emissions characterization were consistent with the target concentrations for each stove on study days.

Study Completion/Missing Data

Of the 26 participants who missed study sessions, 12 missed only one session. Approximately half of the missed study sessions were due to scheduling conflicts that arose after a participant enrolled in the study; one quarter were due to illnesses on scheduled study dates, and one quarter were due to the participants being enrolled in the study late, after the rest of their sequence cohort had completed the first study session.

Four participants withdrew from the study prior to completing six study sessions. Additionally, errors with data logging and our exposure chamber operation resulted in the loss of data relevant to single sessions that were not repeated for five participants. Finally, one participant completed six study sessions, however after the first session, we switched from left-side measurements to right-side to accommodate a medical implant in their left arm; their first session was therefore censored from the dataset. More study sessions were missing from the LPG, gasifier, and fan rocket treatments (44 of the 48 participants completed these treatments) than other treatments (rocket elbow: 45, three stone fire and control: 47). Within the sessions completed, 11 individual data points are missing. Reasons for missing individual data points included scheduling

conflicts for participants that resulted in them leaving a study day without completing the three-hour or 24-hour follow-up time point or a data recording error that resulted in loss of data.

Age, sex, and BMI were comparable between participants who missed sessions and those who did not. For the 22 participants who completed all treatments within sequence, 55% were male, average BMI was $23.4 \pm 2.3 \text{ kg/m}^2$ (range 19.4 to 28.7), and average age was 28.4 ± 4.0 years (range 20.7 to 36.1). For the 26 participants who completed all treatments within sequence, 54% were male, average BMI was $23.4 \pm 2.2 \text{ kg/m}^2$ (range 19.4 to 27.6), and average age was 26.9 ± 3.1 years (range 21.9 to 34.1).

Health Measurement Timing

Baseline pre-exposure measurements occurred on average 25 minutes before entering the exposure facility (range 66 to 12 min). The three sets of post-treatment health measurements were scheduled to start immediately post-exposure, three hours post exposure, and 24 hours post-exposure. The average times of blood pressure measurements were 30 minutes (range: 16 to 49 min; standard deviation: 4.2 minutes), 3 hours and 26 minutes (range: 3 hours 12 min to 3 hours 47 min; standard deviation: 4.8 minutes), and 24 hours and 13 minutes (range 22 hours 13 min to 25 hours 44 min; standard deviation: 30 minutes) post-exposure. Individuals were consistent with themselves across sessions: the average difference in time of day at which each individual's health data were collected was 10 minutes, 11 minutes, and 38 minutes for immediately, three hours, and 24 hours post-exposure, respectively.

Potential Confounders

Alcohol, Caffeine, and Medication Use

Reported alcohol consumption, caffeine intake, and medication use among participants was low (see Tables S2 and S3). Univariate models did not find statistically significant associations between alcohol use, medication use, or caffeine use and treatment type (not shown).

Mode of Commute

Participants were asked to use the same mode of commute into the facility on each study day. Driving was the most common mode of commute (59% of all trips to the facility for the first study day and 56% of all trips for the second study day involved a car), followed by bike (36% of all trips on the first study day and 31% on the second study day; see Tables S4 and S5). Twenty-five of the 48 participants did not change their habits regarding first day commute mode across the six study sessions (i.e., consistently reported the exact same mode for all six sessions). Twenty-two of the 48 participants did not change their habits regarding the second day commute mode use across the six study sessions.

Sleep Quantity

Most participants reported getting an “average” amount of sleep (self-defined “average” as the typical hours of sleep per night; 74% for the night before the study session began and 75% for the night before the second study day); the amount of people reporting below-average sleep was less for the second study day than the first (19% for the night before the study session began vs. 10% for the night before the second study day; see Tables S6 and S7). Only 11 of the 48 participants did not change their habits regarding sleep prior to the start of a study session across the six study sessions (i.e., consistently reported the exact same sleep levels for all six sessions). Thirteen of the 48 participants did not change their habits regarding sleep the night before the 24-hour measurements across the six study sessions.

Ambient PM_{2.5}

Mean ambient PM_{2.5} (measured outdoors at a monitoring site approximately 1 mile from the study facility) in the 24-hours prior to the start of a study day ranged from 4.9 µg/m³ (control) to 9.8 µg/m³ (fan rocket; see Table S8). Minimum recorded mean PM_{2.5} was 0.9 µg/m³ (three stone fire) and maximum recorded mean PM_{2.5} was 17.6 µg/m³ (fan rocket). Ambient PM_{2.5} was higher for all treatments compared to the control except the gasifier. However, the range of ambient PM_{2.5} overall was narrow. We ran a model that was the same as the main model but with an additional covariate of 24-hour average ambient PM_{2.5}; results were comparable to the main model (See Figure S2).

Additionally, we considered whether the rolling average for the 24 hours prior to a blood pressure measurement (e.g., the 24 hours prior to a baseline measurement, 24 hours prior to the 3-hour post-exposure measurement) was associated with that measurement. We ran a linear mixed effect model that estimated the change in systolic blood pressure (in mmHg) for an interquartile range (IQR) change in average ambient PM_{2.5} 24 hours prior to the blood pressure measurement, accounting for correlation between within-person measurements of blood pressure with a random person effect, day of week and season using a random date effect, and average ambient temperature 24 hours prior to the blood pressure measurement. Ambient PM_{2.5} was not meaningfully associated with blood pressure at any measurement time point (see Table S9).

Ambient CO

Mean ambient CO in the 24-hours prior to the start of a study day ranged from 0.25 (rocket elbow) to 0.35 ppm (three stone fire; see Table S10). Minimum recorded mean CO was 0.13 ppm (LPG) and maximum recorded mean CO was 0.70 ppm (three stone fire). Ambient CO was higher for the LPG and three stone fire treatments compared to the control. However, the range of ambient CO overall was determined to be narrow enough to exclude this variable in the main model. We

also considered average CO levels for the six hours prior to the start of a study day; results were similar.

Ambient Temperature

Mean temperature in the 24-hours prior to the start of a study day ranged from 6.0 °C (43 °F; three stone fire) to 15.9 °C (61°F; fan rocket). Minimum recorded mean temperature was -8.5 °C (17 °F; rocket elbow) and maximum recorded mean temperature was 24.2 °C (76 °F; rocket elbow; see Table S11). Temperature was higher for the LPG, fan rocket, and rocket elbow treatments compared to the control. However, the range of temperatures overall was determined to be narrow enough to exclude this variable in the main model.

Fully-Adjusted Model

None of the variables explored appeared to be confounders based on limited variation across the population and/or no association with the treatment type. However, we ran a multivariable model that was equivalent to our main model (containing a random person effect, random date effect, baseline blood pressure term, and categorical treatment) but additionally included variables for alcohol use, caffeine use, medication use, sleep quantity, ambient PM_{2.5}, and ambient temperature. For models estimating blood pressure immediately and 3-hours post-exposure, we used the value of the binary variables that was reported for the 24 hours prior to the baseline measurement; for the 24-hour post-exposure measurement, we used the value reported for the time between the 3-hour post-exposure measurement and the 24-hour post-exposure measurement. For PM_{2.5}, and ambient temperature, we used the rolling average for the 24 hours prior to the measurement. Results of the model indicated that none of the added variables were

significant predictors for blood pressure; further, inclusion of the variables in the model did not meaningfully change the main effect estimates for treatment type (see Figure S3).

Alternative Models

Baseline blood pressures were lowest for the fan-rocket treatment (115.0/68.2 mmHg) followed by the control (115.2/68.6 mmHg) and highest for the three stone fire (117.0/70.2 mmHg). As such, inclusion of a baseline term in the model is justified.

Model with more structured study design parameters, in sequence data only (no makeups)

We developed a mixed-effect model that considered more structured study design parameters relevant to our Williams square, such as each individual's assigned sequence group and the day of week (Monday vs. Wednesday), and only included data that was collected within the intended sequence. Results of this model for systolic pressure indicate no statistical significance ($p < 0.05$) for the various fixed effect "design" terms (day of week, sequence group, or the sequence/day interaction term) (Table S12; Figure S4). There are no differences in main effect estimates compared to the main model with all data (Table S12; Figure S4).

Main model, in sequence only (no makeups)

We ran the main model but on a data set that excluded data collected outside of the intended treatment sequence. Results of this model for systolic pressure indicate no differences in main effect estimates compared to the main model with all data (Table S12; Figure S4).

Main model, remove when exposure value outside narrow range of target

We ran the main model excluding data from study sessions where the exposure mean was outside of a narrowed range around the target value. The narrowed ranges were:

Control: less than 5 $\mu\text{g}/\text{m}^3$ (n = 45; 2 removed)

LPG: 5-15 $\mu\text{g}/\text{m}^3$ (n = 37; 7 removed)

Gasifier: 20-60 $\mu\text{g}/\text{m}^3$ (n = 40; 4 removed)

Fan rocket: 75-125 $\mu\text{g}/\text{m}^3$ (n = 44; 0 removed)

Rocket elbow: 175-300 $\mu\text{g}/\text{m}^3$ (n = 45; 0 removed)

Three stone fire: 350-600 $\mu\text{g}/\text{m}^3$ (n = 47; 0 removed)

Results indicated no considerable differences between the estimates for the treatment effects between this model and the main model (see Figure S5).

Table S1. Characterization of exposure treatments: average emissions.

	control	LPG	gasifier	fan rocket	rocket elbow	three stone fire
Number of tests	2	2	2	2	3	2
PM _{2.5} (µg/m ³ ; gravimetric)	2	10	44	69	196	345
PM _{2.5} (µg/m ³ ; time-resolved)	2	12	50	214	171	445
particle number, 10-500 nm* (/cm ³)	935	17,707	15,255	54,383	69,388	104,739
particle number, 10-100 nm* (/cm ³)	584	17,254	8,925	38,433	39,427	61,404
particle number, 10-30 nm* (/cm ³)	117	15,267	3,504	12,065	9,193	4,435
particle number, 10-50 nm* (/cm ³)	304	16,620	6,156	23,783	21,985	20,550
Percent of total particle number <100 nm*	62%	97%	59%	71%	57%	59%
Percent of total particle number <50 nm*	33%	94%	40%	44%	32%	20%
EC (µg/m ³)	-1 [†]	3	29	38	94	30
OC (µg/m ³)	6	3	8	18	43	132
EC:PM _{2.5} ratio	0.0	0.3	0.7	0.5	0.5	0.1
acetaldehyde (µg/m ³)	18	32	23	19	24	62
acetone (µg/m ³)	34	47	40	33	70	75
formaldehyde(µg/m ³)	10	44	15	24	28	50
hexaldehyde (µg/m ³)	3	6	4	9	10	21
propionaldehyde (µg/m ³)	31	52	34	35	46	53
all carbonyls (µg/m ³)	107	197	128	131	194	293
NO (ppb)	1	4	2	24	24	4
NO ₂ (ppb)	8	10	9	9	10	12
benzene (ppbv)	1	1	6	7	19	35
toluene (ppbv)	4	3	4	5	7	14
ethylbenzene (ppbv)	0	0	0	0	1	2
xylene [‡] (ppbv)	2	1	3	3	4	5

*electrical mobility diameter

[†]Negative value due to small positive EC mass measured on one artifact filter

[‡]Sum of o-xylene and m,p-xylene

Table S2. Alcohol, caffeine, and medication intake by treatments: 24 hours before session start.

Variable	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone	Total	% of total
Total responses	47	44	44	44	45	47	271	
Consumed alcohol	1	1	1	1	3	1	8	3
Consumed caffeine	4	4	4	5	5	2	24	9
Used medications*	4	7	6	5	7	8	37	14
Exposed to smoke	0	1	0	0	1	0	2	1

*This includes some use of daily medications that were approved by the study physician, such as oral contraceptives.

Table S3. Alcohol caffeine, and medication intake by treatments: during the study session.

Variable	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone	Total[†]	% of total
Total responses	46	44	42	43	43	47	266	
Consumed alcohol	2	1	0	0	0	0	3	1
Consumed caffeine	2	4	2	1	2	2	13	5
Used medications*	5	4	8	6	4	7	34	13
Exposed to smoke	0	0	1	2	0	0	3	1

*This includes some use of daily medications that were approved by the study physician, such as oral contraceptives.

[†]Total is lower than in table S2 because some participants missed the 24-hour follow-up period.

Table S4. Mode of commute to facility by treatments: before session start.

Mode	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone	Total
Bike	15	16	17	16	18	14	96
Bike+walk	0	1	1	0	1	0	3
Bus	0	0	0	0	0	0	0
Bus+walk	0	0	1	0	1	0	2
Car	27	24	23	25	24	29	153
Car+walk	3	1	1	1	0	2	8
Walk	2	2	1	1	1	1	8
Not applicable*	0	0	0	1	0	1	2
total	47	44	44	44	45	47	271

*Not applicable: The participant did not report.

Table S5. Mode of commute to facility by treatments: prior to the 24-hour health measurements.

Mode	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone	Total
Bike	14	15	12	14	14	12	81
Bike+walk	0	0	1	0	1	1	3
Bus	1	1	0	2	0	0	4
Bus+walk	2	2	3	2	3	4	16
Car	26	23	24	23	21	29	147
Car+walk	2	1	1	0	2	0	6
Walk	1	1	0	1	2	1	6
Not applicable*	1	1	3	2	2	0	9
Total	47	44	44	44	45	47	271

*Not applicable: The participant did not report or the participant was not present for the 24-hour measurements.

Table S6. Sleep quantity by treatment: night prior to start of study session.

Sleep	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone	Total
Above average	4	2	4	0	5	4	19
Average	33	36	32	36	32	32	201
Below average	10	6	8	8	8	11	51
Total	47	44	44	44	45	47	271

Table S7. Sleep quality by treatment: night prior to the 24-hour health measurements.

Sleep	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone	Total
Above average	6	7	6	8	6	7	40
Average	35	34	33	31	32	35	200
Below average	5	3	3	4	6	5	26
Not applicable*	1	0	2	1	1	0	5
Total	47	44	44	44	45	47	271

*Not applicable: Participant missed the 24-hour follow-up period/survey.

Table S8. Ambient PM_{2.5} Levels* by Treatment: 24 Hours before Session Start.

PM_{2.5}	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone
mean	4.9	7.2	5.3	9.8	6.4	6.6
min	1.5	2.9	1.0	2.0	2.6	0.9
max	9.8	18.8	11.2	17.6	10.6	12.7

*24-hour average in µg/m³

Table S9. Change in systolic pressure per interquartile range change in ambient PM_{2.5}.

Measurement time point	IQR 24-hour average ambient PM_{2.5}	Change in systolic pressure (mmHg) Estimate (95% CI)
Baseline	4.4	-0.10 (-0.96, 0.75)
Immediately post-exposure	4.5	-0.15 (-1.07, 0.76)
Three hours post-exposure	4.7	-0.45 (-1.42, 0.51)
24 hours post-exposure	5.6	-0.34 (-1.44, 0.76)

Table S10. Ambient CO levels* by treatment: 24 hours before session start.

CO	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone
mean	0.27	0.31	0.28	0.29	0.25	0.35
min	0.16	0.13	0.19	0.17	0.17	0.17
max	0.46	0.48	0.45	0.50	0.44	0.70

*24-hour average in ppm

Table S11. Mean temperature* (°C) by treatment: 24 hours before study session.

Temp	Control	LPG	Gasifier	Fan rocket	Rocket elbow	Three stone
mean	7.0	10.9	8.2	15.9	13.3	6.0
min	-7.3	2.9	0.5	4.6	-8.5	-3.1
max	20.2	23.9	14.0	22.3	24.2	15.3

*24-hour average in °C

Table S12. Comparison of model results for three model options: effect estimates and 95% confidence intervals for all model parameters. For random variables, value is the variance and standard deviation.

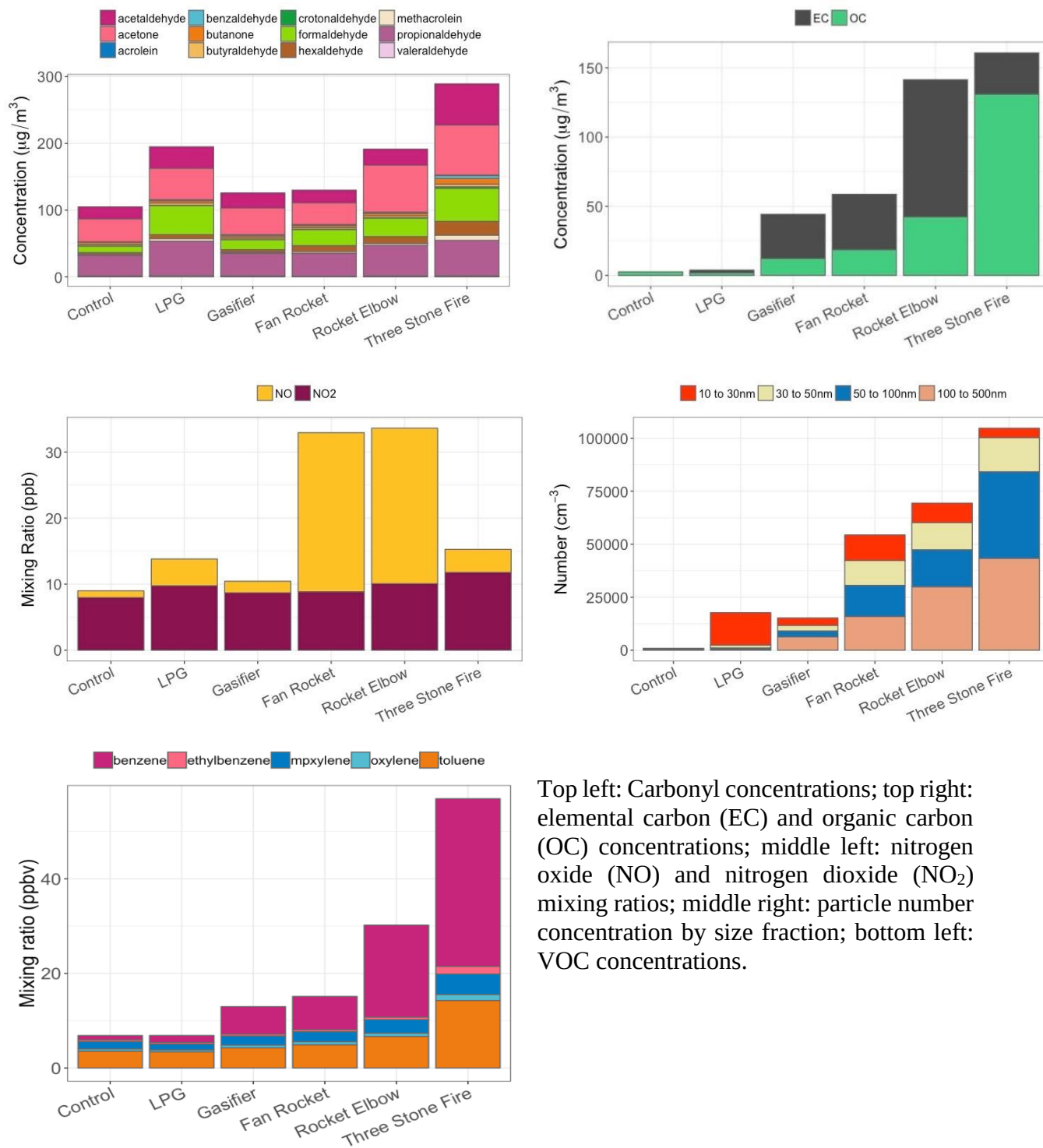
Parameter	MAIN MODEL			MAIN MODEL, IN SEQUENCE DATA			DESIGN MODEL, IN SEQUENCE DATA		
SYSTOLIC PRESSURE									
	immediately post exposure	3 hours post exposure	24 hours post exposure	immediately post exposure	3 hours post exposure	24 hours post exposure	immediately post exposure	3 hours post exposure	24 hours post exposure
n	271	267	264	243	240	240	243	240	240
random date	0.5 (0.7)	0.04 (0.2)	0.0 (0.0)	0.65 (0.8)	0.24 (0.5)	0.0 (0.0)	0.8 (0.9)	0.4 (0.6)	0.0 (0.0)
random person*	26.6 (5.2)	32.6 (5.7)	50.2 (7.1)	29.3 (5.4)	30.8 (5.5)	53.9 (7.3)	32.3 (5.7)	34.9 (5.9)	52.3 (7.3)
random residual	27.3 (5.2)	28.9 (5.4)	26.6 (5.2)	21.9 (4.7)	29.8 (5.5)	22.9 (4.8)	21.2 (4.6)	28.4 (5.3)	22.4 (4.7)
intercept	62.5 (50, 75)	71.5 (58.5, 84.6)	83.1 (69.5, 96.7)	65.3 (53.1, 77.6)	69.7 (56.1, 83.4)	87.8 (74.3, 101.2)	74.1 (60.1, 88.2)	84.1 (68.4, 99.7)	96.0 (80.5, 111.3)
Baseline BP	0.5 (0.4, 0.6)	0.4 (0.3, 0.5)	0.3 (0.2, 0.4)	0.4 (0.3, 0.5)	0.4 (0.28, 0.51)	0.2 (0.1, 0.3)	0.4 (0.3, 0.5)	0.3 (0.2, 0.4)	0.2 (0.1, 0.3)
LPG	-0.2 (-2.5, 2.0)	1.1 (-1.1, 3.3)	3.1 (1.0, 5.3)	0.0 (-2.1, 2.2)	1.6 (-0.8, 4.1)	2.4 (0.3, 4.6)	0.0 (-2.2, 2.2)	1.6 (-0.8, 4.1)	2.3 (0.2, 4.5)
gasifier	-1.81 (-4.0, 0.4)	1.0 (-1.2, 3.2)	2.3 (0.1, 4.5)	-1.0 (-3.3, 1.3)	1 (-1.61, 3.61)	2.2 (-0.1, 4.5)	-1.0 (-3.3, 1.3)	0.95 (-1.6, 3.5)	2.1 (-0.2, 4.4)
fan rocket	-0.4 (-2.7, 1.8)	-1.8 (-4.0, 0.5)	2.5 (0.4, 4.7)	-0.5 (-2.7, 1.7)	-1.35 (-3.9, 1.19)	2.0 (-0.2, 4.2)	-0.6 (-2.8, 1.6)	-1.5 (-4.1, 1.0)	1.9 (-0.3, 4.1)
rocket elbow	-0.6 (-2.8, 1.6)	-0.5 (-2.7, 1.7)	-0.1 (-2.2, 2.1)	-1.3 (-3.5, 0.9)	-0.31 (-2.78, 2.17)	-0.2 (-2.4, 1.9)	-1.3 (-3.4, 0.9)	-0.3 (-2.7, 2.2)	-0.3 (-2.4, 1.9)
three stone	-2.3 (-4.5, -0.1)	-2.1 (-4.3, 0.2)	2.4 (0.3, 4.5)	-1.7 (-3.8, 0.5)	-1.7 (-4.1, 0.7)	2.6 (0.4, 4.7)	-1.6 (-3.7, 0.5)	-1.5 (-3.9, 0.9)	2.6 (0.5, 4.7)
day							-0.0 (-8.6, 8.5)	0 (-9.0, 9.0)	3.4 (-7.2, 14.0)
sequence b							6.4 (-2.1, 14.9)	4.4 (-4.6, 13.3)	8.4 (-2.1, 19.0)
sequence c							-1.8 (-10.2, 6.7)	-4.5 (-13.3, 4.3)	-3.9 (-14.3, 6.6)
sequence d							-1.9	-1.2	0.1

							(-10.3, 6.5)	(-10.0, 7.7)	(-10.3, 10.6)
sequence e							-3.5 (-12.0, 5.0)	-6.8 (-15.7, 2.1)	-2.1 (-12.6, 8.4)
sequence f							-1.4 (-9.9, 7.0)	-4.5 (-13.4, 4.4)	-5.6 (-16.1, 4.9)
day: sequence b							-10.2 (-22.4, 2.0)	-12.0 (-24.9, 0.9)	-15.7 (-30.7, -0.6)
day: sequence c							1.8 (-10.2, 13.8)	4.6 (-8.0, 17.2)	2.7 (-12.2, 17.5)
day: sequence d							-4.0 (-16.2, 8.2)	-6.6 (-19.5, 6.3)	-12.9 (-27.9, 2.2)
day: sequence e							-0.2 (-12.3, 11.8)	2.0 (-10.7, 14.6)	-5.0 (-19.9, 9.8)
day: sequence f							-4.1 (-16.2, 8.0)	-2.5 (-15.2, 10.2)	-3.6 (-18.5, 11.3)
DIASTOLIC PRESSURE									
random date	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
random person*	10.8 (3.3)	12.3 (3.5)	16.7 (4.1)	11.6 (3.4)	11.6 (3.4)	16.6 (4.1)	8.6 (2.9)	10.9 (3.3)	15.9 (4.0)
random residual	13.0 (3.6)	17.4 (4.2)	20.2 (4.5)	12.5 (3.5)	17.8 (4.2)	16.9 (4.0)	12.7 (3.6)	17.8 (4.2)	15.8 (4.0)
intercept	35.3 (28.4, 42.2)	33.7 (25.8, 41.6)	44.8 (36.0, 53.6)	36.7 (30.0, 44.0)	33.6 (25.4, 41.9)	46.2 (37.8, 54.6)	38.1 (30.1, 46.1)	35.6 (26.2, 45.0)	50.0 (40.0, 60)
Baseline BP	0.5 (0.4, 0.6)	0.5 (0.4, 0.6)	0.4 (0.2, 0.5)	0.5 (0.4, 0.6)	0.5 (0.4, 0.6)	0.3 (0.2, 0.5)	0.5 (0.4, 0.6)	0.5 (0.4, 0.6)	0.3 (0.2, 0.4)
LPG	-0.7 (-2.2, 0.8)	-0.0 (-1.7, 1.7)	0.3 (-1.6, 2.2)	-0.4 (-2.0, 1.2)	0.0 (-1.9, 1.9)	-0.6 (-2.4, 1.2)	-0.5 (-2.1, 1.1)	0.02 (-1.9, 1.9)	-0.8 (-2.6, 1.0)
gasifier	-0.8 (-2.2, 0.74)	0.3 (-1.5, 2.0)	-0.4 (-2.3, 1.5)	-0.5 (-2.2, 1.2)	-0.4 (-2.4, 1.6)	-0.8 (-2.7, 1.2)	-0.6 (-2.2, 1.1)	-0.4 (-2.4, 1.6)	-0.8 (-2.7, 1.1)
fan rocket	-0.13 (-1.63, 1.4)	-0.41 (-2.2, 1.3)	-0.1 (-1.9, 1.8)	0.1 (-1.5, 1.7)	-0.4 (-2.3, 1.5)	-1.2 (-3.0, 0.7)	0 (-1.6, 1.6)	-0.5 (-2.4, 1.5)	-1.3 (-3.2, 0.5)
rocket elbow	0.4 (-1.1, 1.8)	0.2 (-1.5, 1.9)	-1.7 (-3.6, 0.2)	0.1 (-1.5, 1.7)	0.1 (-1.9, 2.0)	-2 (-3.8, -0.2)	0.1 (-1.5, 1.7)	0.2 (-1.7, 2.1)	-2.0 (-3.8, -0.2)
three stone	-0.9 (-2.3, 0.6)	-0.8 (-2.5, 0.9)	0.8 (-1.0, 2.7)	-0.7 (-2.3, 0.9)	-0.9 (-2.8, 1.0)	0.5 (-1.3, 2.3)	-0.7 (-2.3, 0.8)	-0.9 (-2.7, 1.0)	0.5 (-1.2, 2.3)
day							-1.5 (-6.2, 3.2)	-0.5 (-5.9, 4.9)	1.2 (-4.9, 7.3)

sequence b							2.1 (-2.6, 6.7)	1.0 (-4.3, 6.2)	4.2 (-1.9, 10.3)
sequence c							-2.9 (-7.5, 1.7)	-3.2 (-8.5, 2.1)	-3.3 (-9.3, 2.7)
sequence d							-4.1 (-8.8, 0.47)	-0.4 (-5.6, 4.9)	-0.8 (-6.8, 5.2)
sequence e							-2.9 (-7.6, 1.7)	-4.0 (-9.3, 1.3)	-0.9 (-6.9, 5.1)
sequence f							-3.5 (-8.2, 1.1)	-1.9 (-7.2, 3.3)	-1.6 (-7.7, 4.4)
day: sequence b							-3.9 (-10.5, 2.7)	-2.4 (-9.9, 5.0)	-7.3 (-15.9, 1.3)
day: sequence c							2.0 (-4.5, 8.6)	4.2 (-3.3, 11.7)	1.4 (-7.1, 10.0)
day: sequence d							1.7 (-5.0, 8.3)	-2.9 (-10.5, 4.7)	-7.7 (-16.3, 0.9)
day: sequence e							-2.4 (-9.0, 4.2)	-1.3 (-8.9, 6.2)	-3.2 (-11.7, 5.4)
day: sequence f							0.34 (-6.3, 6.9)	-0.9 (-8.4, 6.7)	-2.3 (-10.9, 6.2)

*In the main model and main model using in-sequence data only, variable is a random person effect. In the design model, the variable is a random person effect nested within the day and sequence group.

Figure S1. Pollutant characterization in exposure facility, by treatment.



Top left: Carbonyl concentrations; top right: elemental carbon (EC) and organic carbon (OC) concentrations; middle left: nitrogen oxide (NO) and nitrogen dioxide (NO₂) mixing ratios; middle right: particle number concentration by size fraction; bottom left: VOC concentrations.

Figure S2. Effect estimates and 95% confidence intervals for mean difference in systolic pressure (mmHg) for stove treatments compared to control for main model compared to the model including ambient PM_{2.5} variable.

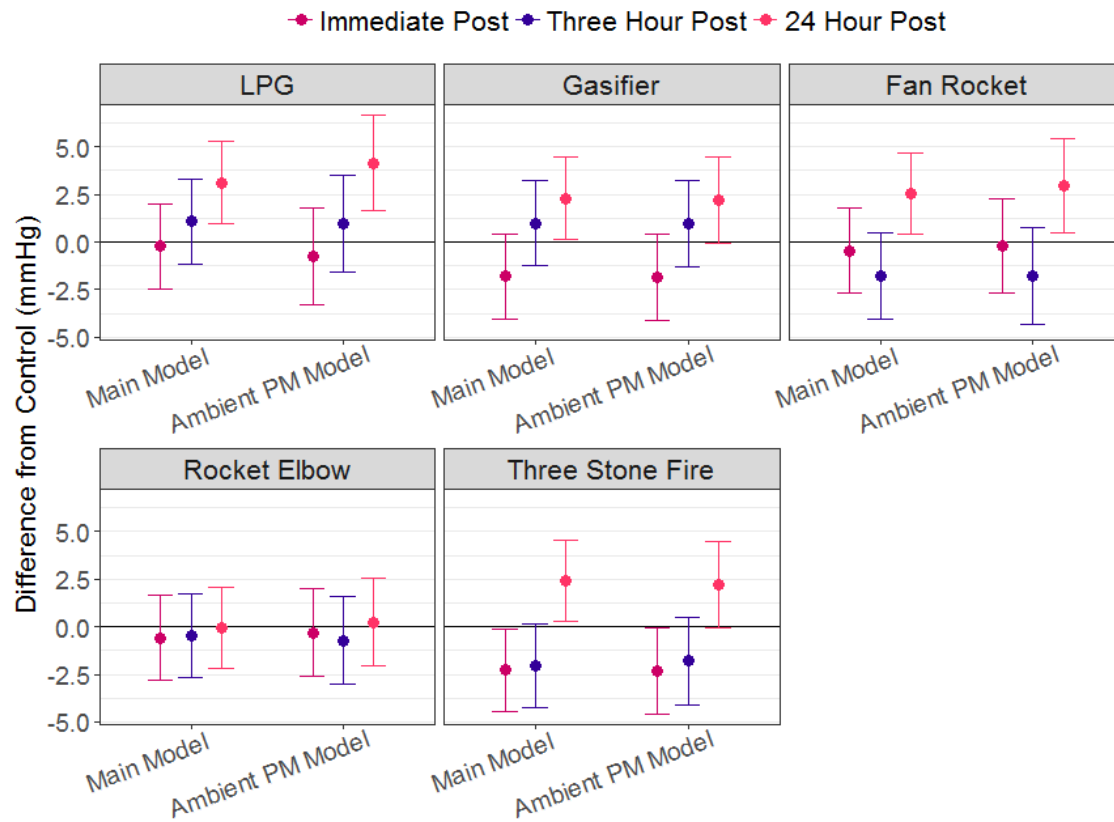


Figure S3. Effect estimates and 95% confidence intervals for mean difference in systolic pressure (mmHg) for stove treatments compared to control for the main model versus the fully adjusted model. Fully adjusted model contains additional variables of alcohol consumption, caffeine consumption, medication use, sleep quantity, ambient PM_{2.5}, and ambient temperature.

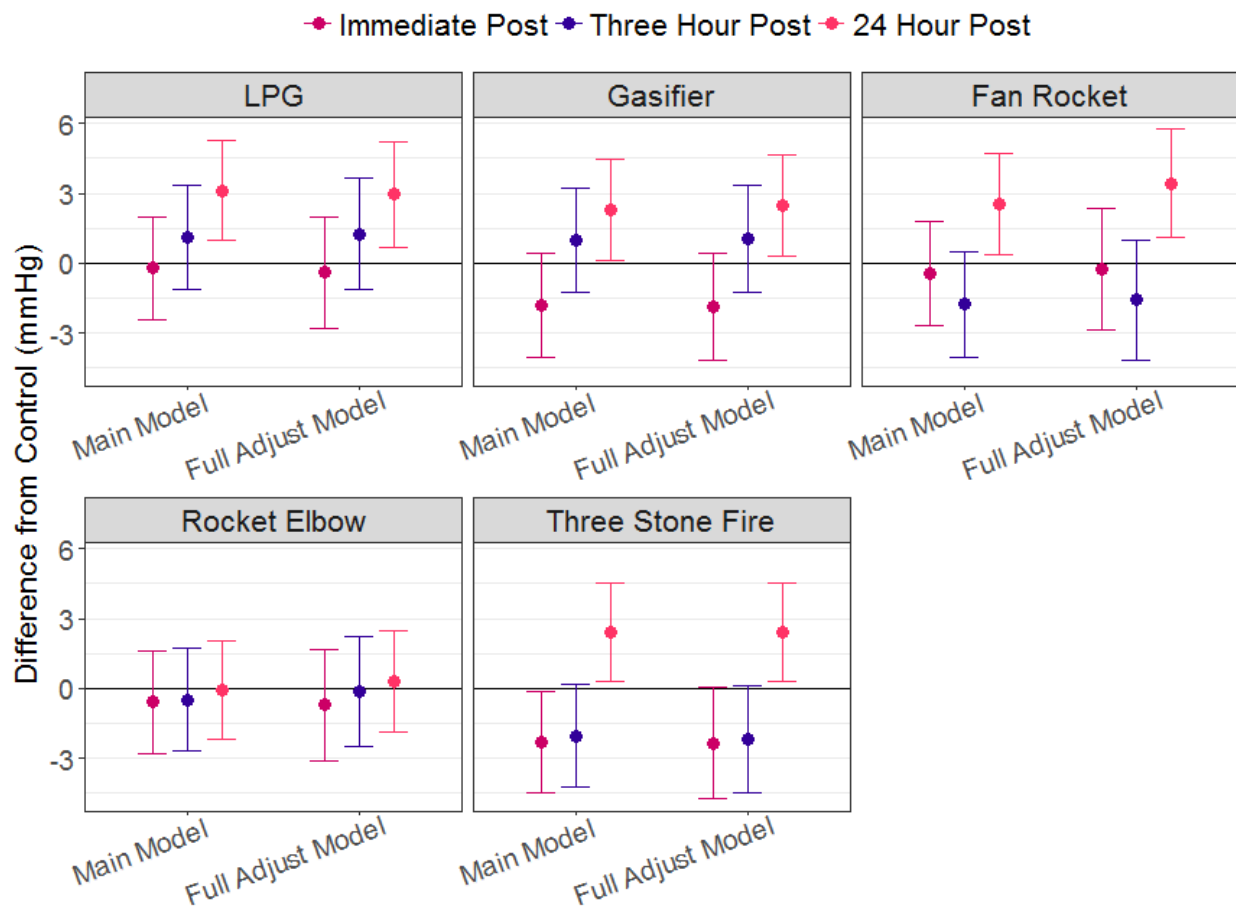


Figure S4. Effect estimates and 95% confidence intervals for mean difference in systolic pressure (mmHg) for stove treatments compared to control for the three model types.

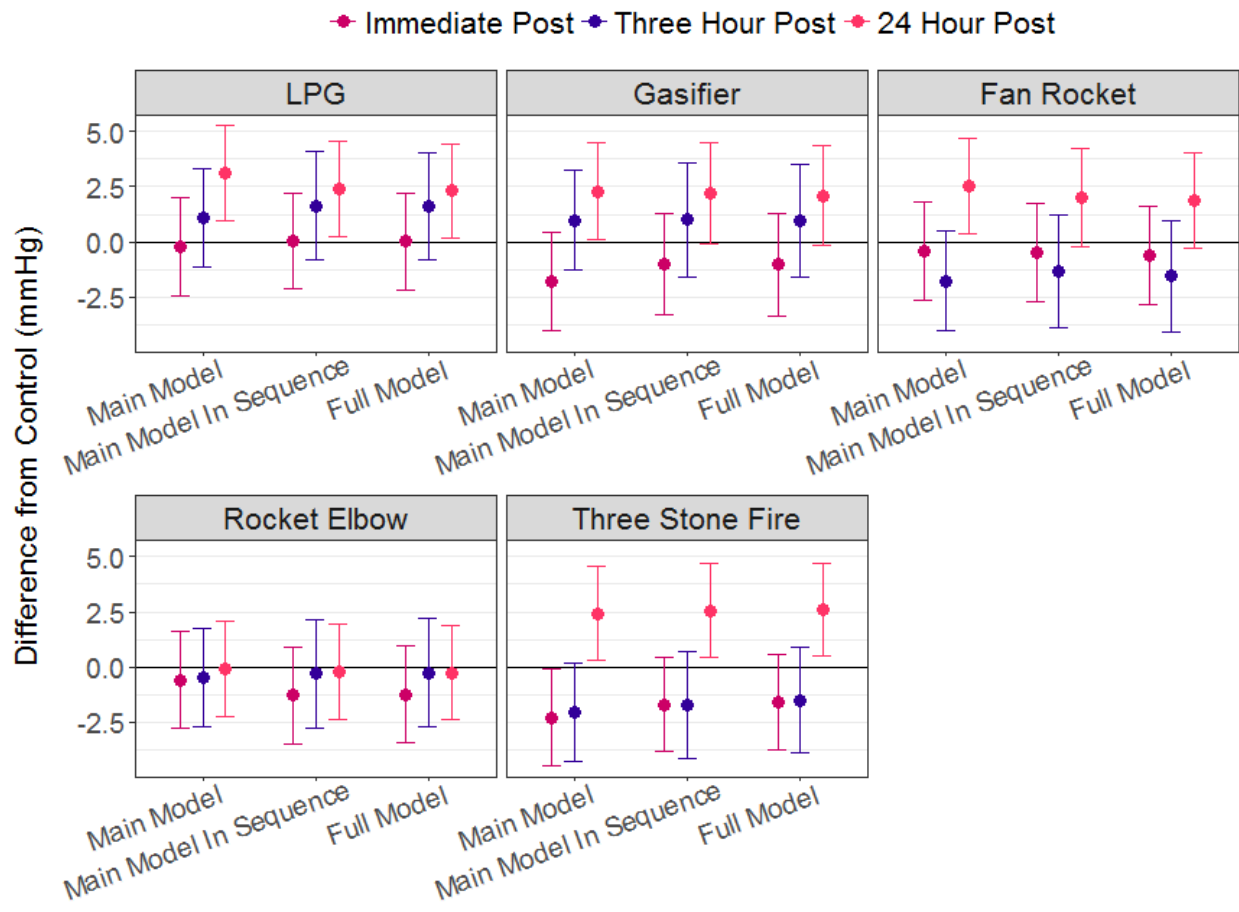


Figure S5. Effect estimates and 95% confidence intervals for mean difference in systolic pressure (mmHg) for stove treatments compared to control: comparison of main model to model with exposure outliers removed.

